

Complex Number 11

1. Simplify as fully as possible

a. $\frac{(\cos \theta + i \sin \theta)^6 (\cos \theta + i \sin \theta)^{-3}}{(\cos \theta - i \sin \theta)^4}$

b. $\frac{(\cos 3\theta + i \sin 3\theta)^5 (\cos 2\theta - i \sin 2\theta)^{-4}}{(\cos 4\theta - i \sin 4\theta)^{-7}}$

2. a. Write $1 + i$ in the form $r(\cos \theta + i \sin \theta)$

b. Hence, or otherwise, find $(1 + i)^{17}$ in the form $a + ib$, where a and b are integers

3. Let $z = 1 + i\sqrt{3}$

a. Express z in mod-arg form

b. Express z^{11} in the form $a + ib$, where a and b are real.

4. Let $z = -\sqrt{3} + i$

a. Find the values of $|z|$ and $\arg z$

b. Hence, or otherwise, show that $z^7 + 64z = 0$

5. a. Express $\sqrt{3} - i$ in mod-arg form.

b. Express $(\sqrt{3} - i)^7$ in mod-arg form.

c. Hence express $(\sqrt{3} - i)^7$ in the form $x + iy$, where x and y are real.

6. a. Express $-1 - i\sqrt{3}$ in mod-arg form.

b. Express $(-1 - i\sqrt{3})^5$ in mod-arg form.

c. Hence express $(-1 - i\sqrt{3})^5$ in the form $x + iy$, where x and y are real.

7. a. i. Use that $z^{10} - 1 = (z^2)^5 - 1$ to factorise $z^{10} - 1$.

ii. Hence or otherwise find the roots of the equation $z^8 + z^6 + z^4 + z^2 + 1 = 0$.

iii. Hence show that $\cos \frac{\pi}{5}$ is a root of the equation $2 \cos 4\theta + 2 \cos 2\theta + 1 = 0$. (You may assume that if $z = \cos \theta + i \sin \theta$ then $z^n + \frac{1}{z^n} = 2 \cos n\theta$)

iv. Hence find the exact value of $\cos \frac{\pi}{5}$.

b. Let w be the root of $z^{10} - 1 = 0$ with the smallest positive argument.

i. Show that $w^2, w^3, w^4, w^6, w^7, w^8$ and w^9 are the other non real roots of $z^{10} - 1 = 0$.

ii. By expanding $(w + w^9)(w^2 + w^8)(w^3 + w^7)(w^4 + w^6)$ and using that w is a root of $z^8 + z^6 + z^4 + z^2 + 1 = 0$ show that $16 \cos^2 \frac{\pi}{5} \cos^2 \frac{2\pi}{5} = 1$.

8. a. If $z = \cos \theta + i \sin \theta$, prove that $z^n + \frac{1}{z^n} = 2 \cos n\theta$

b. Solve $z^8 + 1 = 0$ and use your solutions to resolve $z^8 + 1$ into real quadratic factors.

c. Use your answers to part a. and b. to prove $\cos 4\theta = 8 \left(\cos^2 \theta - \cos^2 \frac{\pi}{8} \right) \left(\cos^2 \theta - \cos^2 \frac{3\pi}{8} \right)$

9. a. Use De Moivre's Theorem to find formulae for: (You may leave your answer in terms of mixed $\sin \theta$ and $\cos \theta$.)

i. $\sin 7\theta$

ii. $\cos 7\theta$

b. Use your answer to part a. to derive the formula for $\tan 7\theta$ in terms of $\tan \theta$.

c. Using your answer to part a. and b. or otherwise, show that $i \tan \frac{\pi}{7}$ is a solution to the equation $x^6 + 21x^4 + 35x^2 + 7 = 0$. (Do NOT solve this equation.)

10. Use De Moivre's Theorem to show that $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$. Hence show that the solutions of the equation $8x^3 - 6x + \sqrt{3} = 0$ are $x = \sin \frac{\pi}{9}$, $x = \sin \frac{2\pi}{9}$, $x = \sin \frac{13\pi}{9}$. Show that $\sin \frac{\pi}{9} + \sin \frac{2\pi}{9} - \sin \frac{4\pi}{9} = 0$.
11. Show that $\frac{1}{2}(1 + i\sqrt{3})(\cos \theta + i \sin \theta) = \cos\left(\frac{\pi}{3} + \theta\right) + i \sin\left(\frac{\pi}{3} + \theta\right)$
12. Simplify $\frac{(\cos 5\theta - i \sin 5\theta)^2 (\cos 3\theta + i \sin 3\theta)^8}{(\cos 2\theta + i \sin 2\theta)^5}$
13. Use De Moivre's Theorem with $n = 2$ to show that: $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$ and $\sin 2\theta = 2 \sin \theta \cos \theta$.
- c. Hence express $\cos \frac{2\pi}{n}$ & $\sin \frac{2\pi}{n}$ in terms of $\cos \frac{\pi}{n}$ & $\sin \frac{\pi}{n}$.
- d. Using the result of part b. to prove that: $\left(1 + \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}\right)^n = -2^n \cos^n \frac{\pi}{n}$
14. Prove the following:
- a. $\sqrt{a + bi} + \sqrt{a - bi} = \sqrt{2(\sqrt{a^2 + b^2} + a)}$
- b. $\sqrt{a + bi} - \sqrt{a - bi} = i \sqrt{2(\sqrt{a^2 + b^2} - a)}$
15. If $x + iy = \sqrt{\frac{a+ib}{c+id}}$, prove that $(x^2 + y^2)^2 = \frac{a^2 + b^2}{c^2 + d^2}$
16. Let $w = \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}$.
- a. Show that w^k , where k is an integer, is a solution of $z^7 - 1 = 0$.
- b. Show that $w^3 + w^2 + w + 1 + w^{-1} + w^{-2} + w^{-3} = 0$.
- c. State the expression for $w^k + w^{-k}$, and hence show that
- i. $\cos \frac{\pi}{7} - \cos \frac{2\pi}{7} + \cos \frac{3\pi}{7} = \frac{1}{2}$.
- ii. $\cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{3\pi}{7} = \frac{1}{8}$.
17. Given $z = \cos \theta + i \sin \theta$,
- a. Use the De Moivre's theorem, to show that $\cos n\theta = \frac{1}{2}(z^n + z^{-n})$
- b. Hence deduce that: $\cos \theta \cos 2\theta = \frac{1}{2}(\cos \theta + \cos 3\theta)$.
18. Given the complex number z , such that $z = k(\cos \theta + i \sin \theta)$, where k is a real number and $0 < \theta < \pi$. Show that: $\arg(z + k) = \frac{1}{2}\theta$.

Answer

1. a. $\text{cis } 7\theta$ b. $\text{cis}(-5\theta)$
2. a. $\sqrt{2}\text{cis} \frac{\pi}{4}$ b. $256 + 256i$
3. a. $2\text{cis} \frac{\pi}{3}$ b. $1024 - 1024\sqrt{3}i$
4. a. $2, \frac{5\pi}{6}$
5. a. $2\text{cis}\left(-\frac{\pi}{6}\right)$ b. $128\text{cis} \frac{5\pi}{6}$ c. $-64\sqrt{3} + 64i$
6. a. $2\text{cis}\left(-\frac{2\pi}{3}\right)$ b. $32\text{cis} \frac{2\pi}{3}$ c. $-16 + 16\sqrt{3}i$
7. a. i. $z^{10} = (z^2 - 1)(z^8 + z^6 + z^4 + z^2 + 1)$ ii. $z = \text{cis}\left(\frac{2k\pi}{10}\right)$, $k = 1, 2, 3, 4, 6, 7, 8, 9$ iv. $\cos \frac{\pi}{5} = \frac{1+\sqrt{5}}{4}$
8. b. $z^8 + 1 = (z^2 - 2\cos \frac{\pi}{8}z + 1)(z^2 - 2\cos \frac{3\pi}{8}z + 1)(z^2 - 2\cos \frac{5\pi}{8}z + 1)(z^2 - 2\cos \frac{7\pi}{8}z + 1)$
9. a. i. $\sin 7\theta = 7\cos^6 \theta \sin \theta - 35\cos^4 \theta \sin^3 \theta + 21\cos^2 \theta \sin^5 \theta - \sin^7 \theta$ ii. $\cos 7\theta = \cos^7 \theta - 21\cos^5 \theta \sin^2 \theta + 35\cos^3 \theta \sin^4 \theta - 7\cos \theta \sin^6 \theta$ b. $\tan 7\theta = \frac{7\tan \theta - 35\tan^3 \theta + 21\tan^5 \theta - \tan^7 \theta}{1 - 21\tan^2 \theta + 35\tan^4 \theta - 7\tan^6 \theta}$
12. $\cos 4\theta + i \sin 4\theta$
13. b. $\cos \frac{2\pi}{n} = \cos^2 \frac{\pi}{n} - \sin^2 \frac{\pi}{n}$, $\sin \frac{2\pi}{n} = 2 \sin \frac{\pi}{n} \cos \frac{\pi}{n}$